Grade 8 Name Answer Key
Unit 12 Date
Lesson 6 Practice Period

For questions 1-3, consider the expression $18w^4m^2 - 27w^5m$.

1. What is the greatest common factor (GCF) of the two terms in the expression?

$9w^4m$

2. Rewrite $18w^4m^2 - 27w^5m$ using the GCF as one of the factors.

$9w^4m(2m - 3w)$

3. Verify using the distributive property that the expression you determined in question 2 is equivalent to the original expression.

$9w^4m(2m - 3w) = 18w^4m^2 - 27w^5m$

For questions 4-7, consider the expression $10a^2b^7 - 30a^5b^4$.

4. What is the greatest common factor (GCF) of the two terms in the expression?

$10a^2b^4$

5. Rewrite $10a^2b^7 - 30a^5b^4$ using the GCF as one of the factors.

$10a^2b^4(b^3 - 3a)$

6. Write an equivalent expression to $10a^2b^7 - 30a^5b^4$ that does not use the GCF as one of the factors. Answers may vary.

$5ab(2ab^6 - 6a^4b^3)$

7. Verify using the distributive property that the expressions you determined in questions 5 and 6 are equivalent to the original expression.

$10a^2b^4(b^3 - 3a^3) = 10a^2b^7 - 30a^5b^4$ ✓

$5ab(2ab^6 - 6a^4b^3) = 10a^2b^7 - 30a^5b^4$ ✓

For questions 8-10, rewrite each expression using the GCF as one of the factors.

8. $-h^7s^3 - 4h^5s^5$

$-h^5s^3(h^2 + 4s^2)$

9. $-5r^4p^7 + 25r^2p^3$

$-5r^2p^3(r^2p^4 - 5)$

10. $4f^3g^6 - 18fg$

$2fg(2f^2g^5 - 9)$